Squeezing-enhanced Homodyne detection between Coherent States in Noisy Communication Channels

Konstantinos Pourgourides

Faculty of Applied Sciences, Delft University of Technology

Internship Supervisor

Professor Ioannis Krikidis

IRIDA Research Center for Communication Technologies, University of Cyprus

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Abstract

In general, non-orthogonal quantum states cannot be discriminated from one another with absolute certainty due to the nature of Quantum Mechanics. In this project, we conduct a numerical study to investigate whether certain modifications on two non-orthogonal quantum states can make them more distinguishable in a detection scenario. More specifically, we focus on the binary phase-shift-keyed (BPSK) protocol, where a transmitter encodes the information in non orthogonal, phase-shifted coherent states $|\pm a\rangle$, while the receiver attempts to discriminate between them using Homodyne detection. In parallel, we also investigate the Helstrom bound, which defines the ultimate theoretical bound of discrimination ability. We modify these states with quadrature squeezing while keeping the number of photons $\langle \hat{n} \rangle$ fixed, amounting to a constant energy budget. Additionally, further modifications affect the states due to noise sources during preparation and transmission, such as phase diffusion and thermal noise. We observe that squeezing enhances the ability of a Homodyne detector to discriminate two coherent states in the noise-free regime, as long as it kept below a critical threshold. Additionally, we show that even in the presence of noisy state preparation and transmission, squeezing can still provide enhancement as long as the noise is also kept under a certain threshold.

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1 Introduction

1.1 The Helstrom Problem: How well can we discriminate between two non-orthogonal quantum states?

Let's assume we have two non-orthogonal quantum states $|\psi\rangle_{\pm}$. The Helstrom problem asks the question: how well can we fundamentally discriminate between these two states? Since they are non orthogonal, one can be expressed in terms of the other

$$|\psi\rangle_{+} = \underbrace{\langle\psi_{+}|\psi_{-}\rangle\,|\psi\rangle_{-}}_{\text{component along } \psi_{-}} + \underbrace{|\psi\rangle_{+}^{\perp}}_{\text{component orthogonal to } \psi_{-}} ; \tag{1}$$

thus, there is a probability $|\langle\psi_{+}|\psi_{-}\rangle|^2$ of failing to correctly identify $|\psi\rangle_{+}$ upon a projective measurement. The larger the overlap between the states, the larger the probability of making an error. For two coherent states $|\pm a\rangle$, which are central to this study, eq. (1) becomes

$$|a\rangle = e^{-2|a|^2}\,|-a\rangle + |a\rangle^{\perp} \ . \tag{2}$$

When the two states have equal prior probability of being selected to for identification, the lowest theoretical error probability is given by the Helstrom bound [1] in eq. (3):

$$P^{\min}(|\psi_{\pm}\rangle) = \frac{1}{2}\left(1 - \sqrt{1 - |\langle\psi_{+}|\psi_{-}\rangle|^2}\right) \ . \tag{3}$$

In the more general case of mixed states represented by density matrices $\rho_{\pm}$, the Helstrom bound is given by

$$P^{\min}(\rho_{\pm}) = \frac{1}{2}\left[1 - \mathrm{Tr}\sqrt{(\rho_{+} - \rho_{-})^2}\right] \ . \tag{4}$$

1.2 The Binary Phase-Shift-Keyed (BPSK) Communication Protocol

In the BPSK communication protocol [2], Alice's task is to encode information (bits 0 and 1) in two quantum states separated by a π-phase-shift and transmit them to Bob. Upon reception, Bob's is responsible for identifying each transmitted state, which will reveal whether the corresponding information bit is 0 or 1.

$$|\psi\rangle = \begin{cases} |a\rangle \Rightarrow \text{bit } 0 \ , \\ |-a\rangle \Rightarrow \text{bit } 1 \ . \end{cases} \tag{5}$$

In the context of continuous-variable (CV) quantum communications, this protocol can be realized with coherent states of light [3, 4], since they are easily generated by commercially available lasers, have their phase properties easily manipulated, and as their name suggests, retain their coherence over long distances [5]. Due to the non-orthogonality of the encoded states, there is a fundamental probability that Bob will make an error even without the realistic obstacles of an experimental method like Homodyne detection. Moreover, various noise sources can affect the states during encoding and transmission [6], which will further damage Bob's discrimination abilities. In this study, we are interested in whether squeezing prior to displacement of the vacuum (displaced squeezed states, as opposed to simple coherent states) can enhance Bob's ability to discriminate the two states under constant energy budget. Under this consideration, eq. (5) is now

$$|\psi\rangle = \begin{cases} |a, r\rangle \Rightarrow \text{bit } 0 \ , \\ |-a, r\rangle \Rightarrow \text{bit } 1 \ . \end{cases} \tag{6}$$

Throughout the study we consider $a, r \in \mathbb{R}^{+}$, meaning that the displacement is only over the x-axis of the phase space, and the squeezed (anti-squeezed) quadrature is $\hat{x}_0 \equiv \hat{x} = \hat{a} + \hat{a}^{\dagger}$ ($\hat{x}_{\pi/2} \equiv \hat{p} = -i(\hat{a} - \hat{a}^{\dagger})$).

1.3 Common Noise Sources in a Communication Channel

1.3.1 Noise-Free Scenario

In the absence of any noise, the coherent states (CS) and their number of photons are given by

$$|\pm a\rangle = \hat{D}(\pm)\,|0\rangle \;\; ; \quad N_{\text{CS}} = |a|^2_{\text{CS}} \; . \tag{7}$$

For displaced squeezed states (DSS), the corresponding relations are

$$|\pm a, r\rangle = \hat{D}(\pm a)\hat{S}(r)\,|0\rangle \;\; ; \quad N_{\text{DSS}} = |a|^2_{\text{DSS}} + \sinh^2 r \; . \tag{8}$$

It is important to note that since $N_{\text{CS}} = N_{\text{DSS}}$ under constant energy conditions, $|a|_{\text{CS}} \geq |a|_{\text{DSS}}$. We additionally define the squeezing fraction $\beta \in [0, 1]$

$$\beta \equiv \frac{N_{\text{squeezing}}}{N_{\text{tot}}} \; , \tag{9}$$

which equals $\sinh^2 r/(\sinh^2 r + |a|^2_{\text{DSS}})$ in the noise-free case. Essentially, it corresponds to the fraction of energy allocated to squeezing, and it will be generalized accordingly in subsequent sections.

1.3.2 Phase Diffusion

Phase diffusion is a process that takes place due to to environmental noise and fluctuation during transmission, resulting in the loss of phase coherence for the state [7]. We can describe its effect through the following differential equation for the state density matrix [4]

$$\Gamma^{-1}\dot{\rho}(t) = 2\hat{n}\rho\hat{n}^\dagger - \hat{n}^\dagger\hat{n}\rho - \rho\hat{n}^\dagger\hat{n} \; ; \tag{10}$$

we may write the density operator in the Fock basis by using the resolution of identity

$$\rho = \mathbb{I}\rho\mathbb{I} = \left(\sum_n |n\rangle\langle n|\right)\rho\left(\sum_m |m\rangle\langle m|\right) = \sum_{n,m}\langle n|\rho|m\rangle\,|n\rangle\langle m| = \sum_{n,m}\rho_{n,m}\,|n\rangle\langle m| \; . \tag{11}$$

Then, eq. (10) becomes

$$\begin{aligned}
&\Gamma^{-1}\dot{\rho}(t) = 2\sum_{n,m} nm\rho_{n,m}\,|n\rangle\langle m| - \sum_{n,m} n^2\rho_{n,m}\,|n\rangle\langle m| - \sum_{n,m} m^2\rho_{n,m}\,|n\rangle\langle m| \; , \\
\Rightarrow &\sum_{n,m}\left[\Gamma^{-1}\dot{\rho}_{n,m} - 2nm\rho_{n,m} + (n^2+m^2)\rho_{n,m}\right]|n\rangle\langle m| = 0 \; , \\
\Rightarrow &\dot{\rho}_{n,m} + \Gamma(n-m)^2\rho_{n,m} = 0 \; .
\end{aligned} \tag{12}$$

Solving the above yields

$$\rho_{n,m}(t) = e^{-(n-m)^2\Gamma t}\rho_{n,m}(0) \; , \tag{13}$$

with the full density matrix being

$$\rho(t) = \sum_{n,m}\rho_{n,m}(t)\,|n\rangle\langle m| = \sum_{n,m} e^{-(n-m)^2\Gamma t}\rho_{n,m}(0)\,|n\rangle\langle m| \; . \tag{14}$$

By the use of Fourier theory, we can rewrite the above equation as

$$\rho(t) = \int_{-\infty}^{+\infty}\left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}}\right)\hat{U}_\phi\rho(0)\hat{U}_\phi^\dagger d\phi = \int_{-\infty}^{+\infty} P_\sigma(\phi)\times(\text{phase-rotated pure state})d\phi \; , \tag{15}$$

where $\hat{U}_\phi = e^{-i\hat{n}\phi}$ and $\sigma \propto \sqrt{\Gamma}$ is the noise parameter. Evidently, the resulting state is a statistical ($\sim \mathcal{N}(0, \sigma^2)$) mixture of the phase-rotated CS/DSS pure states, where

$$\hat{U}(\phi)\rho_{\text{CS}}(0)\hat{U}(\phi) = e^{-i\hat{n}\phi}\hat{D}(\pm a)\left|0\right\rangle\left\langle 0\right|\hat{D}^\dagger(\pm a)e^{i\hat{n}\phi}, \tag{16}$$

$$\hat{U}(\phi)\rho_{\text{DSS}}(0)\hat{U}(\phi) = e^{-i\hat{n}\phi}\hat{D}(\pm a)\hat{S}(r)\left|0\right\rangle\left\langle 0\right|\hat{S}^\dagger(r)\hat{D}^\dagger(\pm a)e^{i\hat{n}\phi}\ . \tag{17}$$

1.3.3 Thermal Noise

Until now, our encoded states were a CS (displaced vacuum $\hat{D}(\pm a)\left|0\right\rangle$) and a DSS (displaced squeezed vacuum $\hat{D}(\pm a)\hat{S}(r)\left|0\right\rangle$ - eq. (17)), where the purity $\mu = \text{Tr}\left[\rho(0)^2\right]$ was unity (pure states). Now, we assume that in the encoding step, instead of a perfect vacuum $\left|0\right\rangle$, we have a thermal state $\nu(\mu)$ of purity $\mu \in [0, 1]$, resulting in a displaced thermal state (DTS) and a displace squeezed thermal state (DSTS) respectively

$$\rho_{\text{DTS}}(0) = \hat{D}(\pm a)\hat{S}(r)\nu(\mu)\hat{S}^\dagger(r)\hat{D}^\dagger(\pm a)\ . \tag{18}$$

$$\rho_{\text{DSTS}}(0) = \hat{D}(\pm a)\hat{S}(r)\nu(\mu)\hat{S}^\dagger(r)\hat{D}^\dagger(\pm a)\ . \tag{19}$$

The thermal state is given by

$$\nu(\mu) = \frac{2\mu}{1+\mu}\sum_{n=0}^{\infty}\left(\frac{1-\mu}{1+\mu}\right)^n\left|n\right\rangle\left\langle n\right|\ , \tag{20}$$

with mean photon number

$$\begin{aligned}
N_{\text{thermal}} = \text{Tr}\left[\nu(\mu)\hat{n}\right] &= \frac{2\mu}{1+\mu}\sum_{n=0}^{\infty}\left(\frac{1-\mu}{1+\mu}\right)^n\text{Tr}\left[\left|n\right\rangle\left\langle n\right|\hat{n}\right] \\
&= \frac{2\mu}{1+\mu}\sum_{n=0}^{\infty}\left(\frac{1-\mu}{1+\mu}\right)^n\underbrace{\sum_m\left\langle m|n\right\rangle\left\langle n\right|\hat{n}\left|m\right\rangle}_{m\delta_{mn}=n} \\
&= \frac{2\mu}{1+\mu}\sum_{n=0}^{\infty}n\left(\frac{1-\mu}{1+\mu}\right)^n \\
&= \frac{1-\mu}{2\mu}\ .
\end{aligned} \tag{21}$$

The general DSTS has mean photon number [4]

$$N_{\text{DSTS}} = |a|^2_{\text{DSTS}} + N\beta + \frac{1-\mu}{2\mu}(1+2N\beta)\ , \tag{22}$$

where we demand that $|a|^2_{\text{DSTS}} \geq 0$, yielding a constraint on μ

$$|a|^2_{\text{DSTS}} = N(1-\beta) + \frac{\mu-1}{2\mu}(1+2N\beta) \geq 0 \tag{23}$$

$$\begin{aligned}
&\Rightarrow \left(\frac{1}{2}+N\beta\right) + N(1-\beta) - \frac{1}{\mu}\left(\frac{1}{2}+N\beta\right) \geq 0 \\
&\Rightarrow \left(\frac{1}{2}+N\beta\right)\left(1-\frac{1}{\mu}\right) \geq N(\beta-1) \\
&\Rightarrow 1-\frac{1}{\mu} \geq \frac{2N(\beta-1)}{1+2N\beta} \\
&\Rightarrow 1-\frac{2N(\beta-1)}{1+2N\beta} = \frac{1+2N}{1+2N\beta} \geq \frac{1}{\mu} \\
&\Rightarrow \mu \geq \frac{1+2N\beta}{1+2N}\ .
\end{aligned} \tag{24}$$

The squeezing fraction β is bounded between 0 and 1 itself; taking this into consideration, μ is constrained in the following way for the lower bound of β

$$\mu \geq \frac{1}{1+2N} \ (\beta = 0); \tag{25}$$

for the upper bound of $\beta = 1$, $\mu \geq 1$. But, since $\mu \in [0, 1]$ as a measure of purity, when all the energy is allocated to squeezing ($\beta = 1$), there is no energy left for thermal noise (or displacement), and thus the state is pure ($\mu = 1$). Taking these further considerations into account, impurity μ is finally bounded in the following way

$$\frac{1}{1+2N} \leq \mu \leq 1. \tag{26}$$

The corresponding results for DTS are derived when no squeezing is involved ($\beta = 0$). The evolution of the state in the general case of phase diffusion is given by eq. (15). Evidently, the resulting state is a statistical ($\sim \mathcal{N}(0, \sigma^2)$) mixture of the phase-rotated DTS/DSTS states, where

$$\hat{U}(\phi)\rho_{\text{DTS}}(0)\hat{U}(\phi) = e^{-i\hat{n}\phi}\hat{D}(\pm a)\nu(\mu)\hat{D}^{\dagger}(\pm a)e^{i\hat{n}\phi}, \tag{27}$$

$$\hat{U}(\phi)\rho_{\text{DSTS}}(0)\hat{U}(\phi) = e^{-i\hat{n}\phi}\hat{D}(\pm a)\hat{S}(r)\nu(\mu)\hat{S}^{\dagger}(r)\hat{D}^{\dagger}(\pm a)e^{i\hat{n}\phi} \ . \tag{28}$$

1.4 Homodyne Detection

Upon receiving the state, Bob employs Homodyne detection [8] to identify whether the received information corresponds to bit 0 or 1. The experimental process of Homodyne detection is illustrated in fig. 1

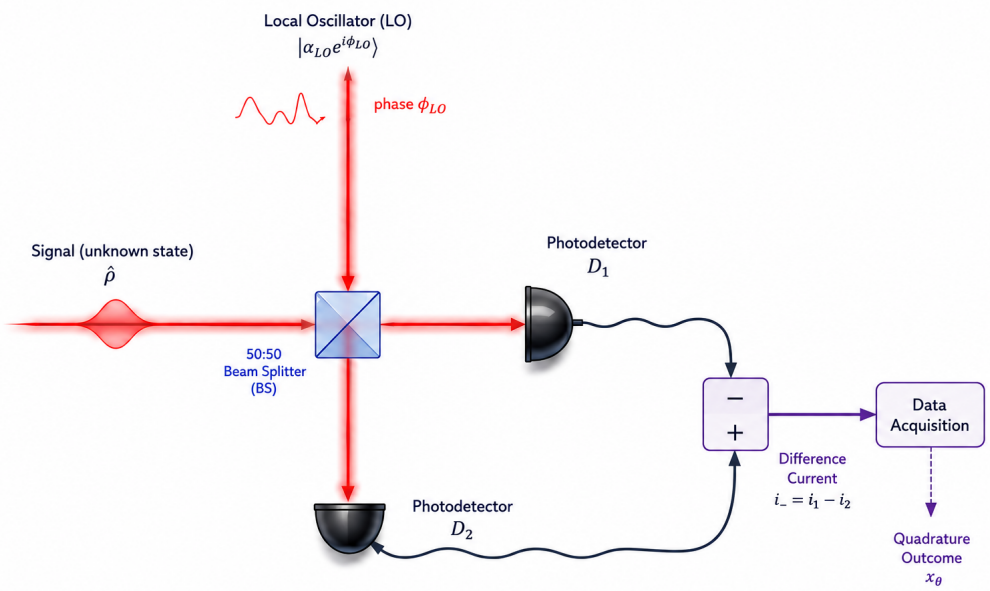


Figure 1: Experimental process of Homodyne detection: the unknown state (signal) is combined with a local oscillator of a known phase θ in a 50-50 beamsplitter. Then, the output current difference between the two photodetectors is used to measure the quadrature $\langle \hat{x}_\theta \rangle$. This illustration was created with ChatGPT.

Bob's measurements of the quadrature $\hat{x}$ are be distributed according to one of two homodyne measurement probability distributions (HMPD), corresponding to 0 or 1. Bob's ability to discriminate between 0 and 1 depends on the characteristics of the distributions, which in turn depend on the amount of displacement a, squeezing r, and any noise that possibly affected the state. In the absence of noise for CS $|\pm a\rangle$, the two HMPD [4] are given by

$$P_{\text{CS}}^{(\text{HD})}(x|\pm) = |\langle x| \pm a\rangle|^2 = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{\left(x \mp 2\sqrt{N}\right)^2}{2}\right] . \tag{29}$$

To discriminate between the two, Bob must identify to which of the two distributions their measurement truly belongs. Due to the non-orthogonality of the encoded states, their HMPDs partially overlap, rendering this a non-trivial task. A depiction of this situation can be seen at fig. (2)

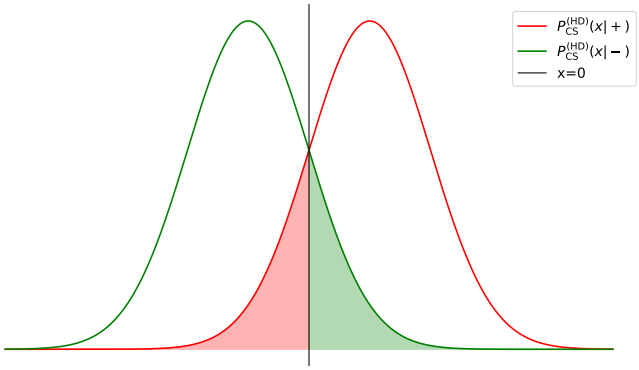


Figure 2: The homodyne measurement probability distributions (HMPDs) for the quadrature measurements x for CS $|\pm a\rangle$ in the absence of noise. Their overlap is a result of their non-orthogonality, which makes the state discrimination process for Bob a non-trivial task.

This overlap hints at the fact that Bob will make errors over multiple measurements. In the next section, we will quantify the error probability of Homodyne detection. The most general formula for the HMPDs in for squeezing fraction β, phase diffusion with noise parameter σ and initial thermal state of purity μ is given by eq. (30)

$$\begin{aligned} P_{DSTS}^{(\mathrm{HD})}(x|\pm) &= \int_{-\infty}^{+\infty} P_\sigma(\phi)\left|\langle x|\hat{D}(\pm a e^{-i\phi})\hat{S}(r)\nu(\mu)\hat{S}^\dagger(r)\hat{D}^\dagger(\pm a e^{-i\phi})|x\rangle\right|^2 d\phi\ , \\ &= \int_{-\infty}^{+\infty} \frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \frac{1}{\sqrt{2\pi V(\phi,\beta,N,\mu)}} \exp\left[-\frac{\left(x \mp 2\sqrt{N(1-\beta)+\frac{\mu-1}{2\mu}(1+2N\beta)}\cos\phi\right)^2}{2V(\phi,\beta,N,\mu)}\right] d\phi\ , \end{aligned} \tag{30}$$

where the variance is

$$V(\phi,\beta,N,\mu) = \frac{\exp\left[-2\sinh^{-1}\left(\sqrt{N\beta}\right)\right]\cos^2\phi + \exp\left[2\sinh^{-1}\left(\sqrt{N\beta}\right)\right]\sin^2\phi}{\mu}\ . \tag{31}$$

To retrieve the distributions without squeezing we take $\beta = 0$. In the absence of phase diffusion and thermal impurities, we have $\sigma = 0,\ \mu = 1$ respectively.

1.4.1 What is the Error Probability of Homodyne Detection?

Bob will attempt to discriminate between the two states with their homodyne measurements x of with the following strategy [4]

$$\begin{cases} x \geq 0 \rightarrow \text{belongs to } \mathrm{P}^{(\mathrm{HD})}(\mathrm{x}|+) \Rightarrow \text{state is } |0\rangle\ , \\ x < 0 \rightarrow \text{belongs to } \mathrm{P}^{(\mathrm{HD})}(\mathrm{x}|-) \Rightarrow \text{state is } |1\rangle\ . \end{cases} \tag{32}$$

The probability that Bob will make an error is proportional to the overlap between the two HMPDs $P^{(\mathrm{HD})}(x|\pm)$, given by

$$P_{\mathrm{err}}^{(\mathrm{HD})} = \frac{1}{2}\left[\int_{-\infty}^{0} P^{(\mathrm{HD})}(x|+)dx + \int_{0}^{+\infty} P^{(\mathrm{HD})}(x|-)dx\right]\ . \tag{33}$$

In the most general case of squeezing fraction β, phase diffusion with noise parameter σ and initial thermal state of purity μ, the homodyne detection error probability (HDEP) is given by

$$P_{\text{err}}^{(\text{HD, DSTS})}(N,\sigma,\mu,\beta) = \frac{1}{2}\int_{-\infty}^{+\infty} \frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \operatorname{erfc}\left[\sqrt{\frac{2\left(N(1-\beta)+\frac{\mu-1}{2\mu}(1+2N\beta)\right)}{V(\phi,\beta,\mu,\nu)}}\cos\phi\right] d\phi \ . \tag{34}$$

Where the variance is the same as in eq. ((31)). To retrieve the HDEP for CS, we take $\beta = 0$. In the absence of phase diffusion and thermal impurities we take $\sigma = 0,\ \mu = 1$ respectively.

1.4.2 Why Squeezing?

One might ask: why do we consider squeezing at all when looking to minimize the HDEP, or equivalently, maximize the discrimination ability of the receiver? The reason to for that is found in the preceding subsections: non-zero squeezing β decreases the variance of the HMPDs (seen in eq. (31)), which in turn decreases the overlap between them; Intuitively, since HDEP is proportional to the overlap, it will be decreased (eq. (33)). A depiction of this statement in the absence of noise is found in fig. (3) when compared to fig. (2).

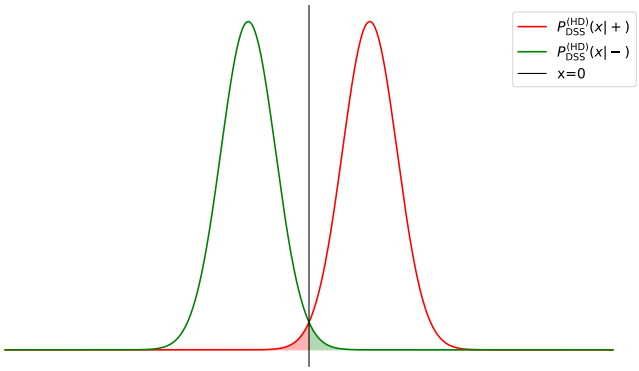


Figure 3: The HMPDs for the quadrature measurements x for DSS $|\pm a, r\rangle$. The use of squeezing has greatly reduced the overlap between the two distributions in comparison to fig. (2), thus leading Bob to make less mistakes when trying to discriminate between the two states with their strategy (eq. (32)).

The nuance is that we are working under a fixed energy budget; hence, we cannot apply indefinite amounts of squeezing to make the distributions infinitely narrower. Additionally, displacement increases the separation between the states, which also helps discrimination (in fact, in the limit $a \to \infty$ the coherent states become exponentially more orthogonal - eq. (2)), meaning we have to balance the allocation of displacement and squeezing to minimize the HDEP. So, we refine our research question as: Which allocation of displacement and squeezing enhances the discrimination ability between two DSS in comparison to CS?

1.5 When does squeezing enhance the discrimination ability?

For the simplest, noise-free case, Bob's ability of discriminating between two CS and DSS becomes equivalent when the following condition holds

$$P_{\text{err}}^{(\text{HD, CS})}(N) = P_{\text{err}}^{(\text{HD, DSS})}(N,\beta) \ . \tag{35}$$

The next important question is whether there are values of β that allow Bob to have better discrimination ability.

1.5.1 Squeezing Threshold - Noise-Free Channels

We will define the squeezing fraction β_{th} as the fraction of squeezing that achieves equal discrimination performance for CS and DSS as in eq. (35)

$$P_{\text{err}}^{(\text{HD, CS})}(N,0) = P_{\text{err}}^{(\text{HD, DSS})}(N,\beta_{\text{th}}) \ . \tag{36}$$

For the noise-free case, there is an analytical solution

$$\beta_{\text{th}}(N) = \frac{4N}{4N+1} \ . \tag{37}$$

Any value of $\beta(N) \in (0, \beta_{\text{th}}(N)]$ achieves better discrimination ability for DSS in comparison to CS. For $\beta(N) \in (\beta_{\text{th}}(N), 1]$, the allocation of energy to displacement is not adequate to separate the two HMPDs enough, and adding more squeezing just makes them narrower but still with significant overlap.

1.5.2 Squeezing Optimum - Noise-Free Channels

We now investigate whether an optimal value of the squeezing fraction, $\beta_{\text{opt}}(N)$, which minimizes the HDEP for DSS, exists. The answer is obtained by the solution of the following equation

$$\left.\frac{\partial P_{\text{err}}^{(\text{HD, DSS})}(N,\beta)}{\partial \beta}\right|_{\beta=\beta_{\text{opt}}} = 0 \ . \tag{38}$$

Noise-free channels admit the following analytical solution

$$\beta_{\text{opt}}(N) = \frac{N}{2N+1} \ . \tag{39}$$

1.5.3 Squeezing Threshold and Optimum - Noisy Channels

In the general case of noisy channels ridden with phase diffusion and thermal noise, the threshold and optimum for the squeezing fraction do not admit analytical solutions. Instead, we have to numerically solve the following equations respectively

$$P_{\text{err}}^{(\text{HD, DTS})}(0 \mid N,\sigma,\mu) - P_{\text{err}}^{(\text{HD, DSTS})}(\beta_{\text{th}} \mid N,\sigma,\mu) = 0 \ , \tag{40}$$

$$\beta_{\text{opt}}(N,\sigma,\mu) = \arg \min_{\text{fi}\in[0,1]} \text{P}_{\text{err}}^{(\text{HD, DSTS})}(\text{fi} \mid \text{N}, \text{œ}, \text{¯}) \ . \tag{41}$$

In the results section, we will see that there is an interplay between the allocation of squeezing and level of noise, making squeezing useful only when the noise parameter σ is smaller than a threshold σ_{th}.

2 Methods

2.1 Monte Carlo Simulation for the Error Probability of Homodyne Detection

To simulate the transmission, reception and Homodyne detection processes in the most general case, we numerically calculate the HDEP $P_{\text{err}}^{(\text{HD})}$ in the following fashion. We create a grid with values for our different parameters:

$$\begin{aligned}
\boldsymbol{N} &= \{0, \ldots, N_{\text{max}}\}, \\
\boldsymbol{\beta} &= \{0, \ldots, 1\}, \\
\boldsymbol{\mu} &= \{(1+2N_{\text{max}})^{-1}, \cdots, 1\}, \\
\boldsymbol{\sigma} &= \{0, \ldots, 1\} \ .
\end{aligned}$$

We proceed by calculating the HDEP (eq. (30)) $P_{\text{err}}^{(\text{HD})}(N,\sigma,\mu,\beta)$, $\forall \sigma \in \boldsymbol{\sigma}$, $\mu \in \boldsymbol{\mu}$, $\beta \in \boldsymbol{\beta}$. Since this is a Gaussian-weighted integral with no analytical solution, we employ the Gauss-Hermite quadrature method [9] (with the substitution $x \equiv \phi/\sqrt{2}\sigma$)

$$P_{\text{err}}^{(\text{HD})}(N,\sigma,\mu,\beta) = \frac{1}{\pi}\int_{-\infty}^{+\infty} e^{-x^2} f(x)\, dx \approx \frac{1}{\pi}\sum_{i}^{n} w_i f(x_i) \ , \tag{42}$$

where (x_i, w_i) are the quadrature points and weights respectively, computed with Python. Then, since $P_{\text{err}}^{(\text{HD})}(N, \sigma, \mu, \beta)$ is the error probability of homodyne measurement, the number of errors K after m state transmissions follows a binomial distribution. The probability of making k errors after m homodyne measurements is

$$P(K = k) = \binom{m}{k} p^k (1-p)^{m-k} \ . \tag{43}$$

Finally, we randomly draw a number of errors K from this distribution, and the estimated HDEP is

$$\hat{P}_{\text{err}}^{(\text{HD})}(N, \sigma, \mu, \beta) = \frac{K}{m} \ . \tag{44}$$

2.2 Numerical Methods for Squeezing Threshold & Optimum

2.3 Threshold

We employ Brent's method [10] to find numerical solutions to equations of the type $f(x) = 0$, just like in eq. (36), where the solution represents the squeezing threshold β_{th}. This approach combines three different methods: Bisection, Secant Method and Inverse Quadratic Interpolation (IQI). It starts with an interval $[a, b]$ which contains the root

$$f(a) f(b) < 0 \ ; \tag{45}$$

in our case, $f(x) = P_{\text{err}}^{(\text{HD, DTS})}(N, \sigma, \mu, 0) - P_{\text{err}}^{(\text{HD, DSTS})}(N, \sigma, \mu, \beta)$, and $[a, b] = [0, 1]$. At each iteration, it tries an interpolation step to estimate the root. If it has three good previous points, Brent estimates the root using a curved interpolation. If IQI isn't suitable, it tries the secant method. Using two points, it estimates the root with a straight line

$$x_{\text{sm}} = b - f(b) \frac{b-a}{f(b) - f(a)} \ . \tag{46}$$

If the interpolation step looks unreliable, it falls back on bisection

$$x_{\text{bis}} = \frac{a+b}{2} \tag{47}$$

After whichever step is chosen, it evaluates $f(x_{\text{new}})$ and updates the interval so that the sign change is preserved.

2.4 Optimum

We employ Brents' minimization algorithm to find the minimum point x^* of a function $f(x)$, just like with the optimal squeezing fraction β_{opt}, being the minimum of $P_{\text{err}}^{(\text{HD})}(\beta \mid N, \mu, \sigma)$ (eq. (41)). It starts with an interval $[a, b]$ which contains the minimum point x^*. Then, it evaluates the function multiple times inside the interval. It proceeds by picking 3 points $(x_i, f(x_i))$, $i = 1, 2, 3$, and fits a parabola $\mathcal{P}(x)$ through them

$$\mathcal{P}(x) = Ax^2 + Bx + C \ . \tag{48}$$

The minimum of the parabola $x_{\text{new}} = -B/2A$ is a potentially good estimate of the minimum. If $x_{\text{new}} \in (a, b)$ and $|x_{\text{new}} - x_{\text{previous}}|$ is not much smaller than $|b - a|$, the point is accepted. If it does not fulfill these criteria, the algorithm employs the golden-section method. It chooses two points $x_1 = b - r(b-a)$ and $x_2 = a + r(b-a)$ where $r^{-1} = \phi = \frac{1+\sqrt{5}}{2} \approx 1.618$, such that they divide the interval $[a, b]$ in the golden ratio. Then,

$$\text{if } \begin{cases} f(x_1) < f(x_2) \rightarrow \text{new interval } [\text{a}, \text{x}_2] \\ f(x_1) > f(x_2) \rightarrow \text{new interval } [\text{x}_1, \text{b}] \end{cases} \ .$$

3 Results

3.1 Noise-Free Scenario

Firstly, we present the HDEP for CS and DSS, $P_{\text{err}}^{(\text{HD, CS})}(N)$ and $P_{\text{err}}^{(\text{HD, DSS})}(N,\beta)$ respectively, under noise-free conditions in fig. (4)

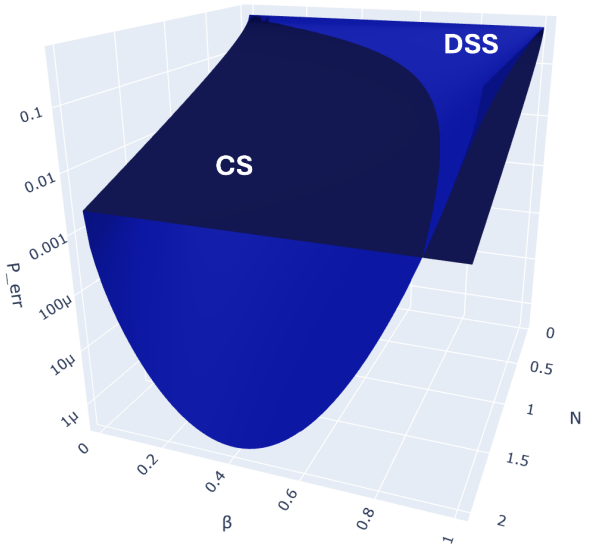


Figure 4: The HDEP planes $P_{\text{err}}^{(\text{HD, CS})}(N)$ and $P_{\text{err}}^{(\text{HD, DSS})}(N,\beta)$ for CS and DSS respectively, under noise-free conditions. The intersection of the surfaces corresponds to $\beta_{\text{th}}(N)$.

Evidently, there is a set of squeezing fractions for which $P_{\text{err}}^{(\text{HD, CS})}(N) = P_{\text{err}}^{(\text{HD, DSS})}(N,\beta)$, representing the threshold of the squeezing fraction $\beta_{\text{th}}(N)$ of eq. (37). Additionally, there is another set of squeezing fractions which minimize $P_{\text{err}}^{(\text{HD, DSS})}(N,\beta)$, representing the optimal squeezing fraction $\beta_{\text{opt}}(N)$ of eq. (39). In the noise-free scenario, these quantities admit analytical solutions which are depicted in fig. (5)

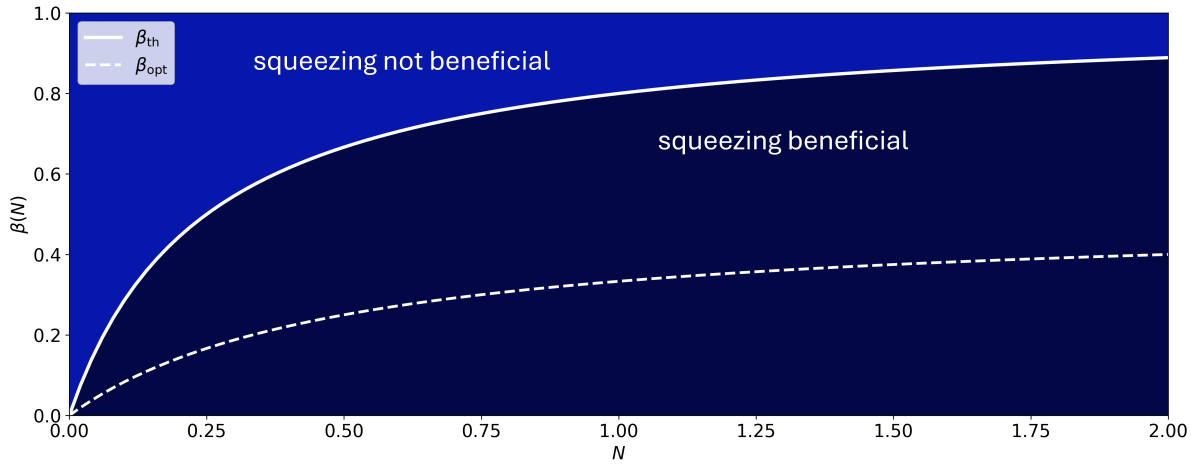


Figure 5: The squeezing fraction threshold $\beta_{\text{th}}(N)$, for which $P_{\text{err}}^{(\text{HD, CS})}(N) = P_{\text{err}}^{(\text{HD, DSS})}(N,\beta_{\text{th}}(N))$, and optimum $\beta_{\text{opt}}(N)$ for which $P_{\text{err}}^{(\text{HD, DSS})}(N,\beta_{\text{opt}}(N))$ is minimum.

Lastly, we present the behavior of the HDEPs as a function of N for both CS and DSS with respect to the Helstrom bound in fig. (6). We observe that in the limit $N \to 0$, there is not enough energy provided by displacement to separate the HMPDs, and thus the discrimination ability is poor in both cases (Helstrom bound and $P_{\text{err}}^{(\text{HD, CS/DSS})} \approx 1$). For intermediate values of N, $P_{\text{err}}^{(\text{HD, DSS})}(N|\beta_{\text{opt}})$ is both lower than $P_{\text{err}}^{(\text{HD, CS})}(N)$ and approaches the Helstrom bound more closely. In the limit $N \gg 1$, the energy is able to separate the two HMPDs sufficiently enough such that the HDEPs in both cases asymptotically tend to zero, albeit much faster for the DSS.

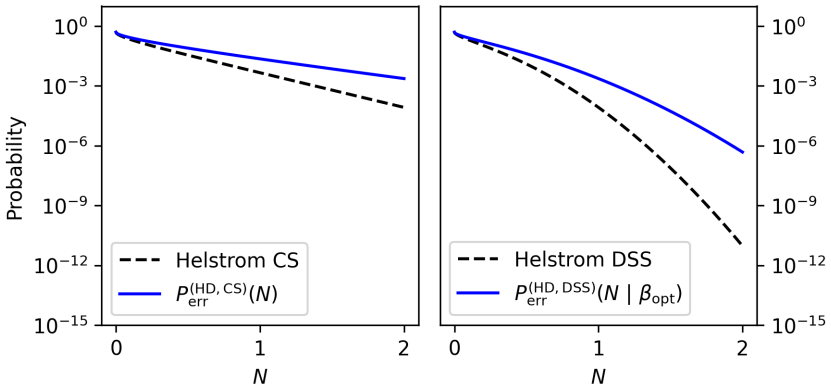


Figure 6: The Helstrom bound and HDEPs $P_{err}^{(HD,\ DSS)}(N|\beta_{opt})$ and $P_{err}^{(HD,\ CS)}(N)$ for CS and DSS respectively.

3.2 Phase Diffusion

We move on to the HDEPs for CS and DSS under phase diffusion for different values of the noise parameter σ, $P_{err}^{(HD,\ CS)}(N|\sigma)$ and $P_{err}^{(HD,\ DSS)}(N,\beta|\sigma)$, respectively in fig. (7)

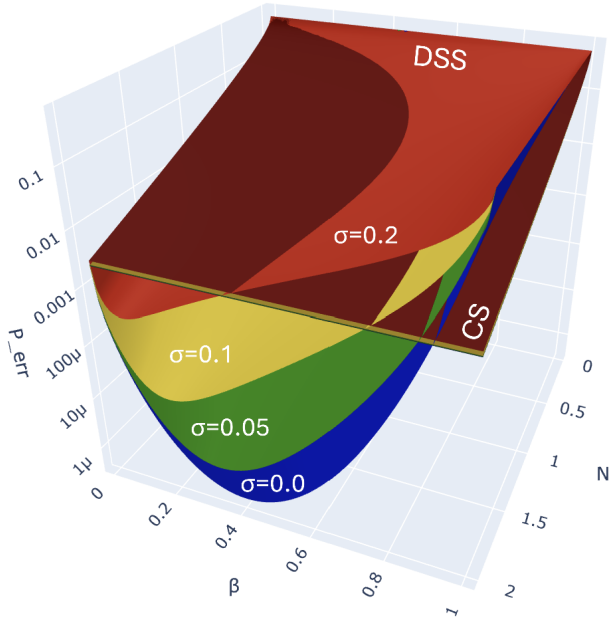


Figure 7: The HDEP planes $P_{err}^{(HD,\ CS)}(N|\sigma)$ and $P_{err}^{(HD,\ DSS)}(N,\beta|\sigma)$ for CS and DSS respectively, under phase diffusion, for different values of the noise parameter $\sigma = \{0.0, 0.05, 0.1, 0.2\}$. The intersection of the surfaces corresponds to $\beta_{th}(N|\sigma)$.

We observe that that as the noise parameter σ moves from lower to higher values, the HDEP planes are displaced upwards, indicating larger error probabilities. Also, when comparing to the noise-free scenario in fig. (4), the intersection of the planes, and thus $\beta_{th}(N|\sigma)$, changes with the noise parameter. Consequently, we proceed by investigating how the squeezing fraction and optimum $\beta_{th/opt}(N|\sigma)$ behave now, in fig. (8). It is clear that as the noise parameter σ is increased, the squeezing fraction threshold and optimum $\beta_{th/opt}(N|\sigma)$ are lowered, meaning the available squeezing values for which DSS outperform CS are fewer. It is also interesting to study the behavior of the squeezing fraction threshold $\beta_{th}(\sigma|N)$ as a function of the noise parameter σ for constant values of photon number N; the corresponding results are shown in fig. (9)

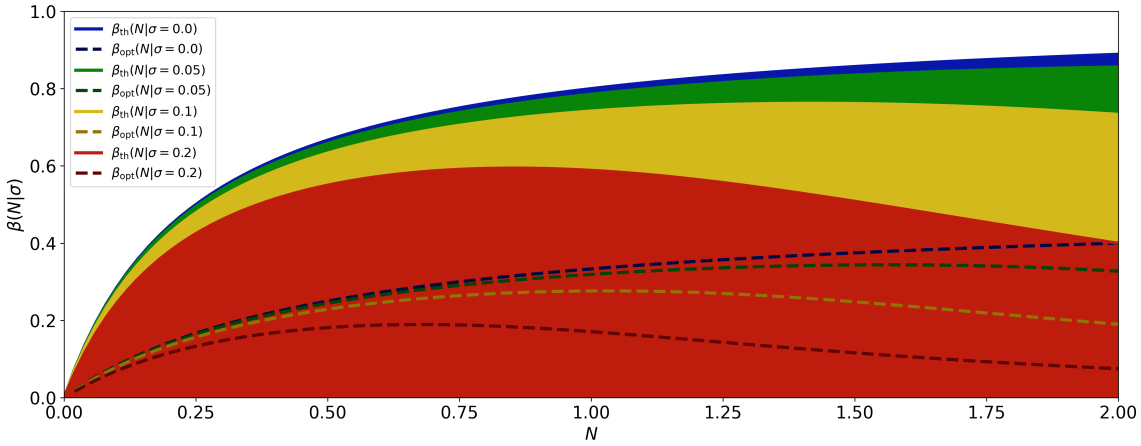


Figure 8: The squeezing fraction threshold $\beta_{\text{th}}(N|\sigma)$, for which $P_{\text{err}}^{(\text{HD, CS})}(N|\sigma) = P_{\text{err}}^{(\text{HD, DSS})}(N, \beta_{\text{th}}(N)|\sigma)$, and optimum $\beta_{\text{opt}}(N|\sigma)$ for which $P_{\text{err}}^{(\text{HD, DSS})}(N, \beta_{\text{opt}}(N)|\sigma)$ is minimum.

It is evident that for each number of photons N, squeezing can still improve state discrimination ability when applied to the vacuum prior to displacement; this time, not only when it is below a certain threshold, but also when, simultaneously, the noise parameter σ does not exceed a certain value. The values of (β, σ) for which DSS outperform CS correspond to the shaded regions in fig. (9), for each photon number N.

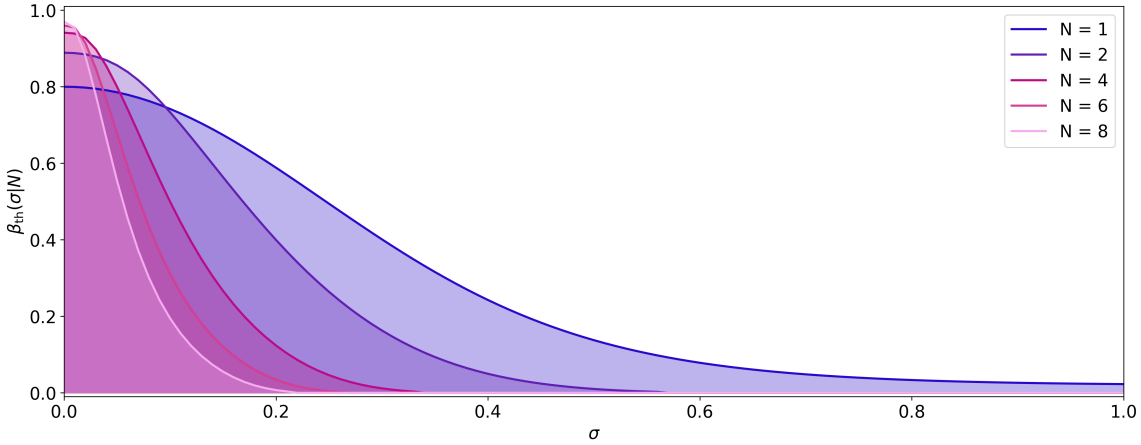


Figure 9: The squeezing fraction threshold $\beta_{\text{th}}(\sigma|N)$ as a function of the noise parameter σ for constant values of photon number N.

Lastly, we present the behavior of the HDEPs as a function of σ and for photon numbers $N = 1, 2$ for both CS and DSS, with respect to the Helstrom bound in fig. (10). We observe that for small values of the noise parameter σ, the HDEP of the DSS can out-perform the Helstrom bound of the CS. Additionally, in the same regime, the same is true for the HDEPs. An interesting observation which ties back to fig. (9), is that for large σ, where squeezing is no longer beneficial, the two HDEPs and Helstrom bounds converge.

3.3 Thermal Noise

Lastly, we present the HDEPs for DTS and DSTS under phase diffusion and thermal noise for noise parameters $\sigma = \{0.0, 1.0\}$ and photon number $N = 2$, $P_{\text{err}}^{(\text{HD, DTS})}(\mu|N, \sigma)$ and $P_{\text{err}}^{(\text{HD, DSTS})}(\mu, \beta|N, \sigma)$ respectively, in fig. (11). We observe that in both cases, higher purity corresponds to lower HDEP. As the noise parameter σ grows, the DSTS error probability plane approaches that of DTS, finally reaching it at a certain value, above which squeezing no further enhances discrimination ability. This is evident from the fact that for $\sigma = 0.0$, the DSTS plane has regions below the DTS plane, while that is not the case for $\sigma = 1.0$.

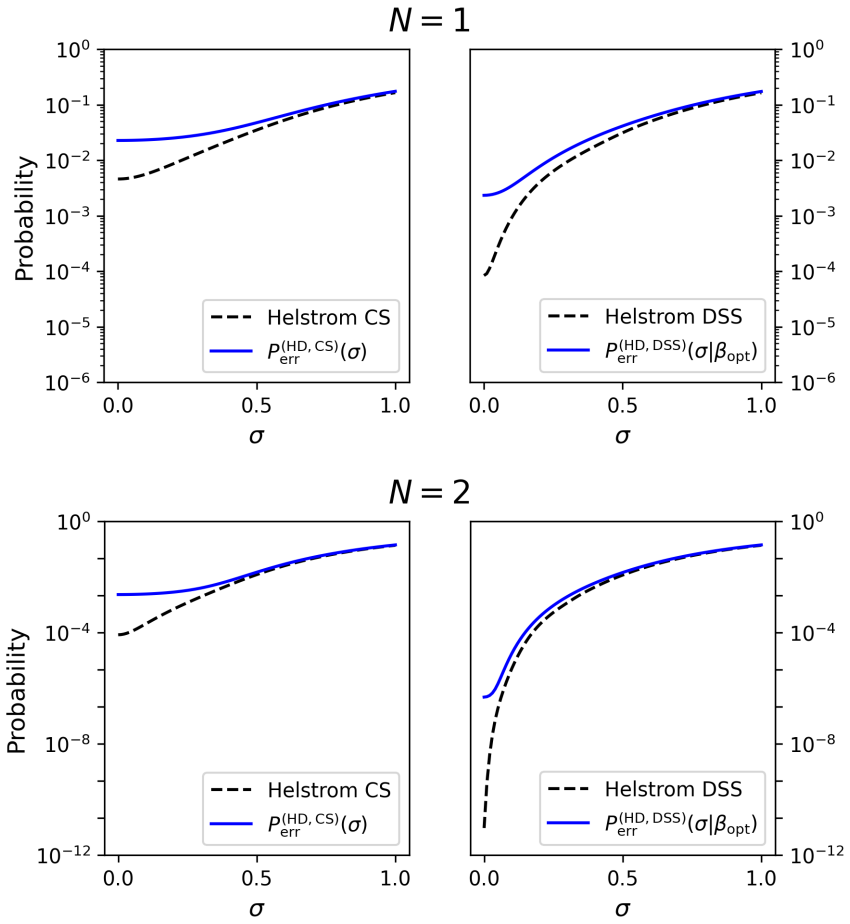


Figure 10: The Helstrom bound and HDEPs $P_{err}^{(HD,\ DSS)}(\sigma|\beta_{opt})$ and $P_{err}^{(HD,\ CS)}(N)$ for photon numbers $N = 1, 2$, for CS and DSS respectively.

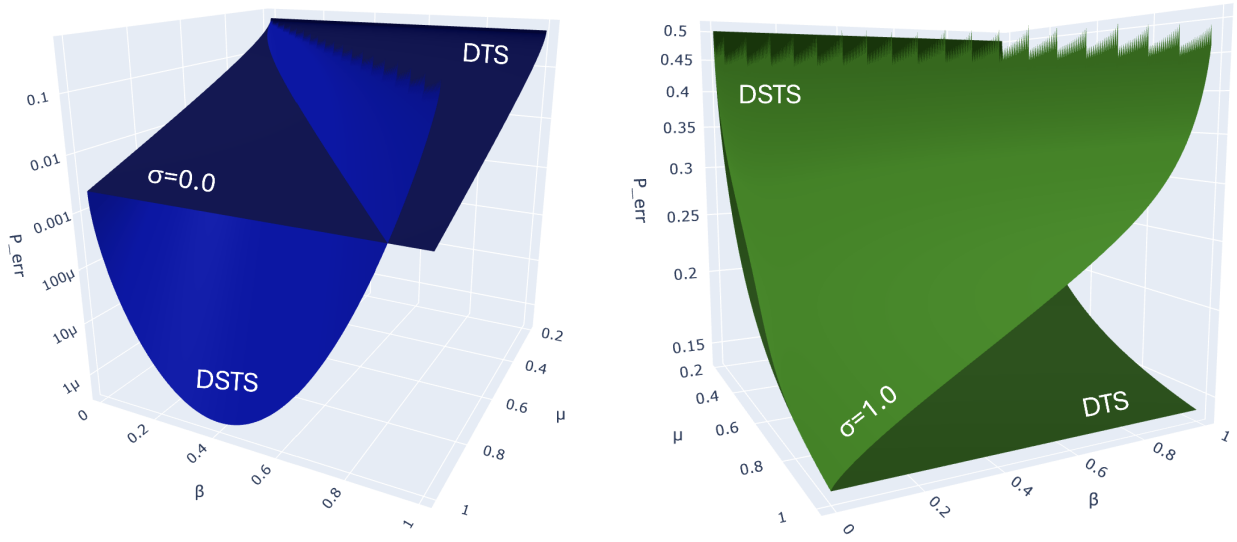


Figure 11: The HDEP planes $P_{err}^{(HD,\ DTS)}(\mu|N,\sigma)$ and $P_{err}^{(HD,\ DSTS)}(\mu,\beta|N,\sigma)$ for DTS and DSTS respectively, under phase diffusion and thermal noise, for different values of the noise parameter $\sigma = \{0.0, 1.0\}$ and photon number $N = 2$. The intersection of the surfaces corresponds to $\beta_{th}(N|\sigma)$.

It is also interesting to study the behavior of the squeezing fraction threshold $\beta_{th}(\mu|N,\sigma)$ as a function of the purity μ for a constant value of photon number N; the corresponding results are shown in fig. (12)

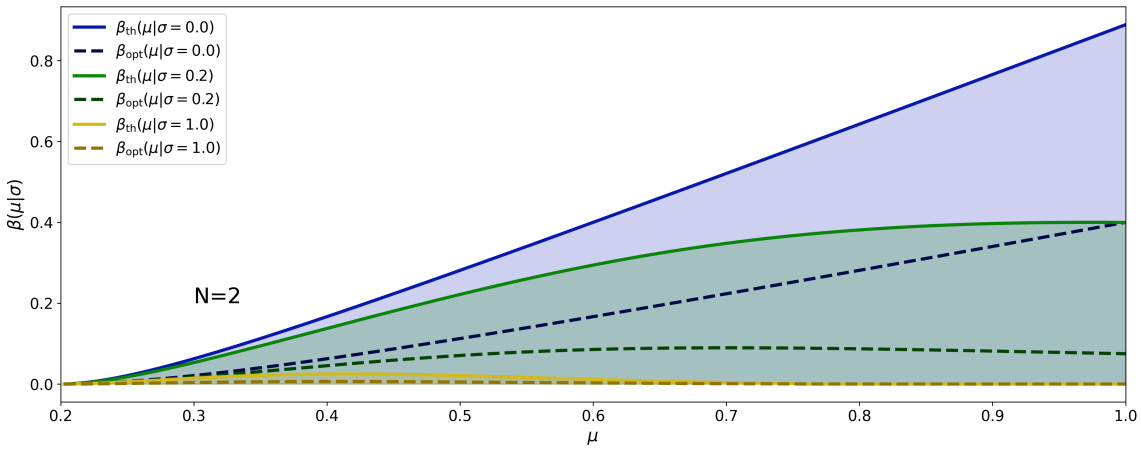


Figure 12: The squeezing fraction threshold $\beta_{\text{th}}(\mu|N,\sigma)$, for which $P_{\text{err}}^{(\text{HD, DTS})}(\mu|N,\sigma) = P_{\text{err}}^{(\text{HD, DSTS})}(\mu,\beta_{\text{th}}(\mu)|N,\sigma)$, and optimum $\beta_{\text{opt}}(\mu|N,\sigma)$ for which $P_{\text{err}}^{(\text{HD, DSTS})}(\mu,\beta_{\text{opt}}(\mu)|N,\sigma)$ is minimum, for photon number $N=2$.

As the noise parameter σ increases, the squeezing fraction threshold is decreased as a function of purity μ. In the large noise regime, there is a value of σ for each N and μ, over which squeezing is no longer beneficial for state discrimination, and $\beta_{\text{opt}}(\mu|N,\sigma_{\text{th}})=0$. This value of σ is referred to as $\sigma_{\text{th}}(\mu|N)$. This noise parameter threshold is depicted in fig. (13).

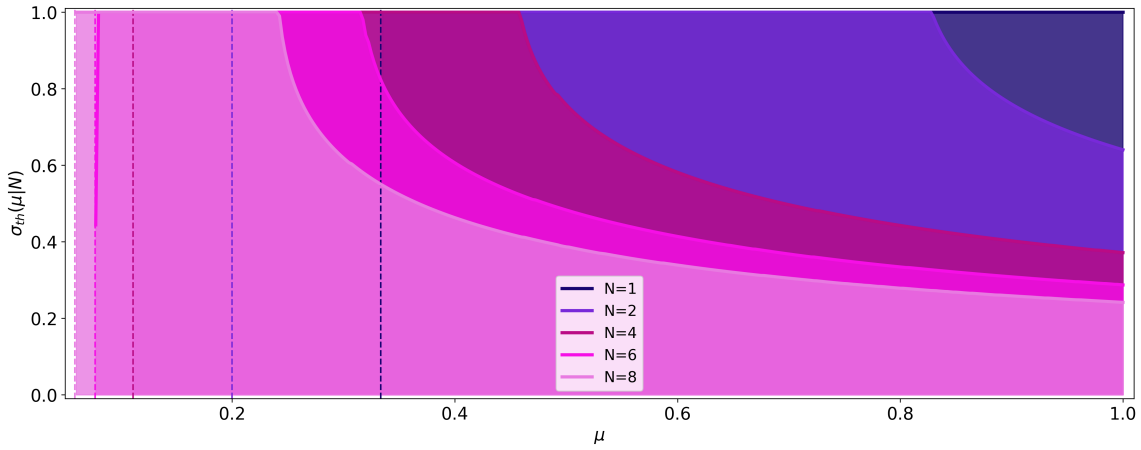


Figure 13: The noise parameter threshold $\sigma_{\text{th}}(\mu|N)$ as a function of the purity μ for constant photon numbers $N=\{1,2,4,6\}$. Squeezing benefits discrimination ability only when the noise parameter is below the indicated threshold at each purity. The dashed lines correspond to the minimum allowed value of purity $\mu=(1+2N)^{-1}$.

Lastly, we present the behavior of the HDEPs a function of μ, for photon number $N=2$ and noise parameter $\sigma=\{0.0,1.0\}$, for both DTS and DSTS with respect to the Helstrom bound, in fig. (14). We observe that in the low noise regime, where the noise threshold has not yet been reached, squeezing is beneficial and both the Helstrom bound and HDEP are significantly smaller for DSTS in comparison to DTS, especially as purity grows. In the high noise regime, where the noise threshold has been crossed (see fig. (13)), squeezing is no longer beneficial and the Helstrom bound and HDEP for both states are approximately the same.

4 Conclusions

In conclusion, we investigated whether the utilization of squeezing in BPSK quantum communication with coherent states can enhance the discrimination performance of homodyne detection under a fixed energy budget, noisy state preparation and transmission. More specifically, the encoded states where two π-phase-shifted coherent states $|\pm a\rangle$, and we considered the noise sources of phase diffusion and thermal noise during state transmission and preparation, respectively. In parallel, we also commented on the theoretically ultimate discrimination performance, the Helstrom bound, and its behavior with respect to the Homodyne detection performance.

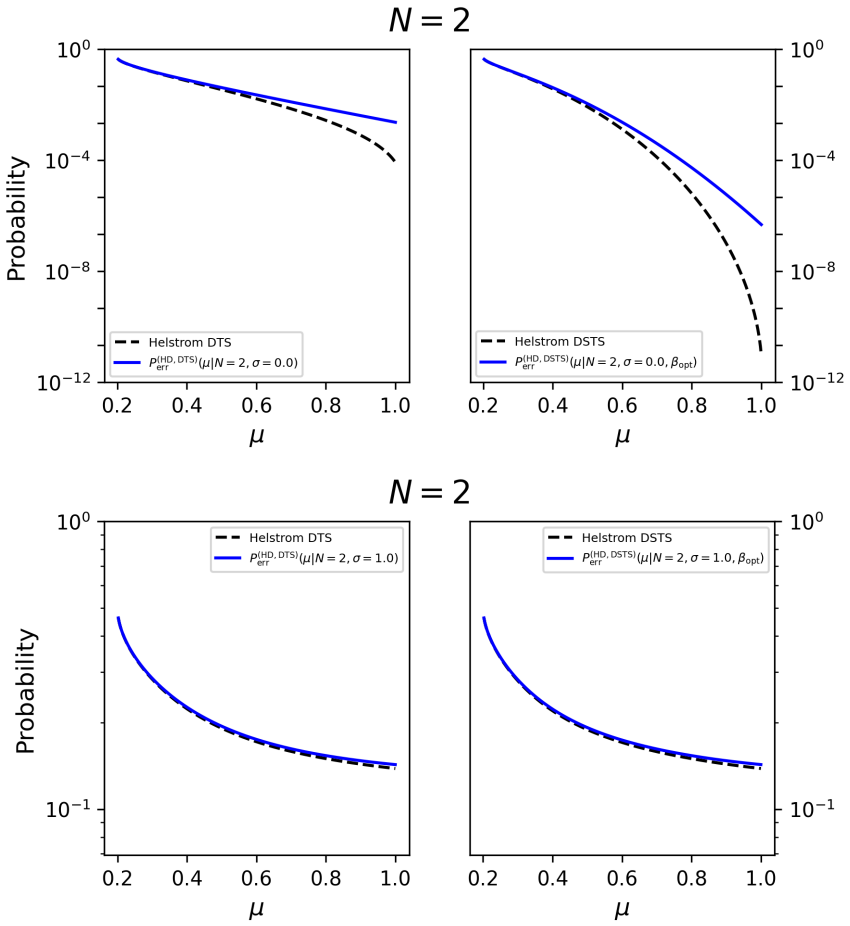


Figure 14: The Helstrom bound and HDEPs $P_{\text{err}}^{(\text{HD, DSTS})}(\mu|N,\sigma,\beta_{\text{opt}})$ and $P_{\text{err}}^{(\text{HD, DTS})}(\mu|N,\sigma)$ for photon number $N = 2$ and $\sigma = \{0.0, 1.0\}$, for DSTS and DTS states respectively.

In the absence of noise, squeezing is beneficial as long as it is kept under a certain threshold $\beta_{\text{th}}(N)$. Additionally, there exists an optimal amount of squeezing $\beta_{\text{opt}}(N)$ which maximizes discrimination performance with homodyne detectors. The Helstrom bound and homodyne detection error probability of displaced squeezed states are significantly lower than those of coherent states with increasing photon number, and both reach zero asymptotically in the limit $N \to \infty$.

In the case of phase diffusion during transmission, we observed that the squeezing threshold and optimum are also a function of the noise parameter σ, meaning that squeezing is not beneficial if the noise is sufficiently large. In the low noise regime ($\sigma \to 0$), the Helstrom bound and homodyne detection error probability for displaced squeezed states are significantly lower than those of coherent states, but as the noise increases ($\sigma \to 1$) they converge, since then squeezing is no longer beneficial.

Lastly, in the case where the seed state (vacuum) is a thermal state of purity μ, the previous observations re-emerged, with the addition that the threshold of the squeezing $\beta_{\text{th}}(\mu|\sigma)$ and noise $\sigma_{\text{th}}(\mu|N)$ are now functions of the purity. The previous observations also hold for the Helstrom bound; displaced squeezed thermal states outperform displaced thermal states in the pure ($\mu \to 1$) regime, only when noise is kept under the corresponding threshold.

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